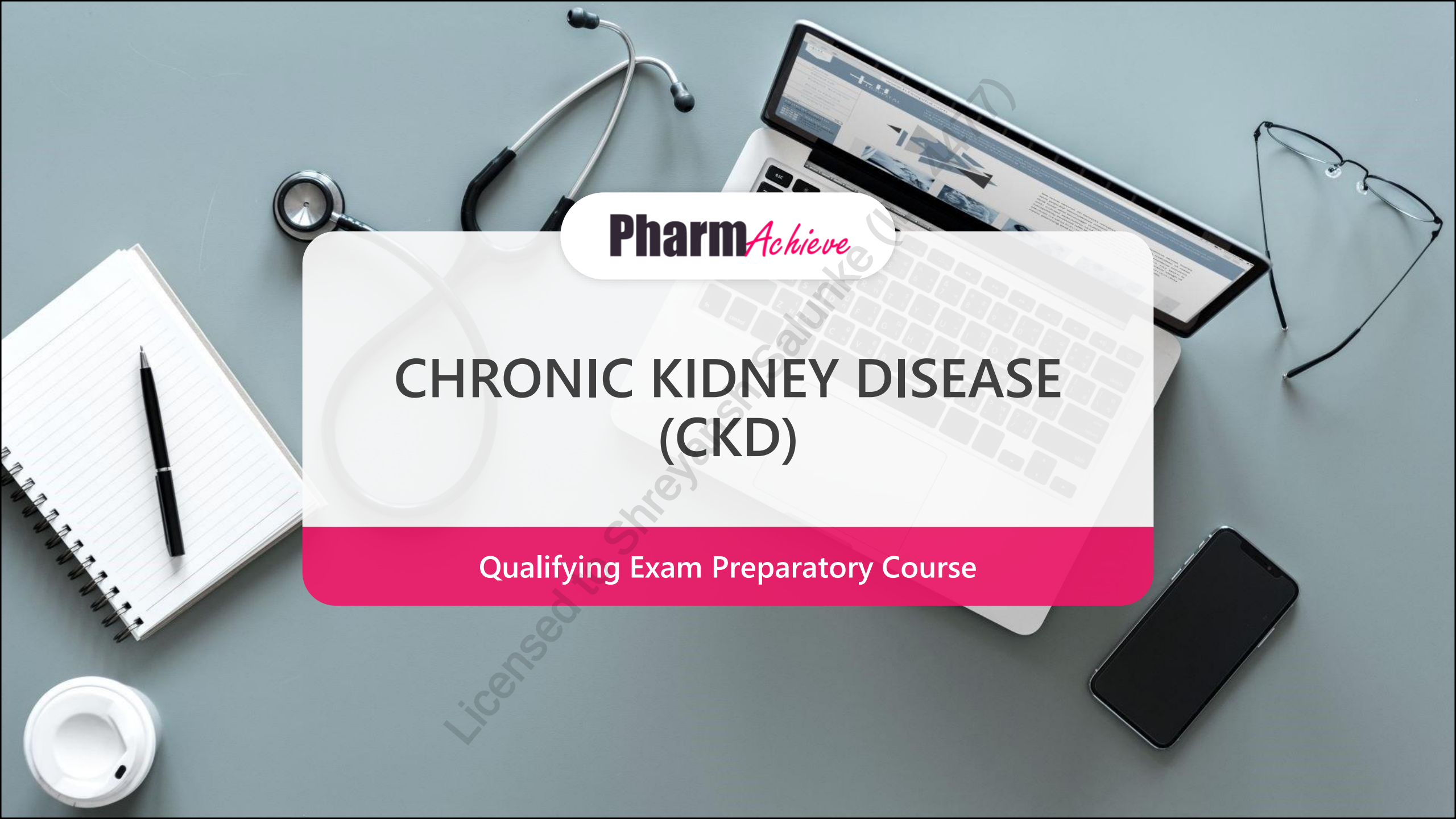


A top-down view of a desk with various items: a stethoscope, a laptop displaying a medical website, a tablet, a pair of glasses, a smartphone, a spiral notebook with a pen, and a coffee cup. A large white pill-shaped graphic is centered over the laptop, containing the course title and name. A faint watermark 'Licensed to Shreyas Salunke' is visible diagonally across the image.

**Pharm***Achieve*

# CHRONIC KIDNEY DISEASE (CKD)

Qualifying Exam Preparatory Course

# COMPETENCIES COVERED IN THIS PRESENTATION...

**Providing Care:  
Clinical Care**

**Providing Care:  
Distribution**

**Knowledge  
& Expertise**

**Communication  
& Collaboration**

**Leadership &  
Stewardship**

**Professionalism**

# LEGEND

|              |                                         |               |                                        |
|--------------|-----------------------------------------|---------------|----------------------------------------|
| <b>ABW</b>   | Actual Body Weight                      | <b>HPT</b>    | Hyperparathyroidism                    |
| <b>adjBW</b> | Adjusted Body Weight                    | <b>HTN</b>    | Hypertension                           |
| <b>ACEI</b>  | Angiotensin Converting Enzyme Inhibitor | <b>IBW</b>    | Ideal Body Weight                      |
| <b>ACR</b>   | Albumin-to-Creatinine Ratio             | <b>IM</b>     | Intramuscular                          |
| <b>ARB</b>   | Angiotensin Receptor Blockers           | <b>IV</b>     | Intravenous                            |
| <b>ATII</b>  | Angiotensin II                          | <b>MDRD</b>   | Modification of Diet in Renal Disease  |
| <b>BMI</b>   | Body Mass Index                         | <b>MI</b>     | Myocardial Infraction                  |
| <b>BG</b>    | Blood Glucose                           | <b>MOA</b>    | Mechanism of Action                    |
| <b>BP</b>    | Blood Pressure                          | <b>NSAIDs</b> | Non-steroidal anti-inflammatory agents |
| <b>BW</b>    | Body Weight                             | <b>PCR</b>    | Polymerase Chain Reaction              |
| <b>CHF</b>   | Congestive Heart Failure                | <b>PE</b>     | pulmonary Embolism                     |
| <b>CKD</b>   | Chronic Kidney Disease                  | <b>PTH</b>    | Parathyroid Hormone                    |
| <b>CrCl</b>  | Creatinine Clearance                    | <b>QOL</b>    | Quality of Life                        |
| <b>CRI</b>   | Chronic Renal Insufficiency             | <b>RBC</b>    | Red Blood Count                        |
| <b>CRRT</b>  | Continuous Renal Replacement Therapy    | <b>SC</b>     | Subcutaneous                           |
| <b>CVD</b>   | Cardiovascular Disease                  | <b>SCR</b>    | Serum Creatinine                       |
| <b>DM</b>    | Disease Management                      | <b>SDOH</b>   | Social Determinants of Health          |
| <b>eGFR</b>  | Estimated Glomerular Filtration Rate    | <b>SLE</b>    | Systemic Lupus Erythematosus           |
| <b>ESA</b>   | Epoetin Stimulating Agents              | <b>SOB</b>    | Shortness of Breath                    |
| <b>ESRD</b>  | End Stage Renal Disease                 | <b>TIBC</b>   | Total Iron Binding Capacity            |
| <b>GFR</b>   | Glomerular Filtration Rate              | <b>Tsat</b>   | Transferrin Saturation                 |
| <b>GI</b>    | Gastrointestinal                        | <b>Tx</b>     | Treatment                              |
| <b>HbA1C</b> | Glycated Hemoglobin                     | <b>T2DM</b>   | Type 2 Diabetes Mellitus               |
|              |                                         | <b>VTE</b>    | Venous Thromboembolism                 |

# DEFINITIONS

## Creatinine Clearance (CrCl)

- Rate of filtration of creatinine by the kidney (marker for GFR)
- Estimate of GFR
- Calculated by Cockcroft-Gault equation (*mg/dL*)

$$\text{CrCl}^* = \frac{[(140 - \text{age}) \times \text{ABW}]}{72 \times \text{SCr}}$$

\*Multiply by 0.85 if female

## Glomerular Filtration Rate (GFR)

OLD

- Total rate of filtration of blood by the kidney
- Use estimated GFR (eGFR) using MDRD formulation
$$\text{GFR}^* = 186.3 \times (\text{SCR})^{-1.154} \times (\text{age in years})^{-0.203}$$
- \*Multiply result by 0.742 if female or 1.212 if of Afro-Caribbean descent

## Calculate IBW:

*Males*

$\text{IBW} = 50 \text{ kg} + 2.3 \text{ kg for each inch over 5 feet}$

*Females*

$\text{IBW} = 45.5 \text{ kg} + 2.3 \text{ kg for each inch over 5 feet}$

Use Adjusted BW if increased BMI:

$\text{adjBW} = \text{IBW} + 0.4 (\text{ABW} - \text{IBW})$

Creatinine and cystatin C–based estimated glomerular filtration rate [eGFR<sub>cr-cys</sub>]

- If cystatin C is available, the GFR category should be estimated from the combination of creatinine and cystatin C

# DEFINITIONS

NEW

## Glomerular Filtration Rate (GFR)

- Total rate of filtration of blood by the kidney
- Use estimated GFR (eGFR) using **CKD-EPI (2021)** Equation  
$$\text{eGFR} = 142 \times \min(\text{Scr}/\kappa, 1)^\alpha \times \max(\text{Scr}/\kappa, 1)^{-1.200} \times 0.9938^{\text{Age}} [\times 1.012 \text{ if female}]^*$$
- More accurate than the MDRD equation, especially in people with higher GFR
- Removing race-based modifiers promotes equitable care for black persons

### Legend and Values:

$S_{\text{cr}}$  (serum creatinine) = mg/dL

$\kappa$  = 0.7 (females) or 0.9 (males)

$\alpha$  = -0.241 (females) or -0.302 (males)

min = indicates the minimum of  $S_{\text{cr}}/\kappa$  or 1

max = indicates the maximum of  $S_{\text{cr}}/\kappa$  or 1

# CHRONIC KIDNEY DISEASE (CKD)

Also known as Chronic Renal Insufficiency (CRI)

- Abnormalities of kidney structure or function present for a minimum of 3 months with Implications for health
  - Progressive loss of structure or function occurring over several months to years, characterized by the gradual replacement of normal kidney structure with interstitial fibrosis
- Classified based on CGA
  - Cause
  - Glomerular filtration rate (GFR) category (G1–G5)
  - Albuminuria category (A1–A3)
- Defined by:

|                                                        |    |                                         |        |                                                                                                                        |
|--------------------------------------------------------|----|-----------------------------------------|--------|------------------------------------------------------------------------------------------------------------------------|
| GFR <60<br>ml/min/1.73 m <sup>2</sup><br>for ≥3 months | OR | ACR ≥30mg/g [3mg/mmol]<br>for ≥3 months | AND/OR | Presence of markers of kidney<br>damage for ≥3 months with or<br>without decreased Glomerular<br>Filtration Rate (GFR) |
|--------------------------------------------------------|----|-----------------------------------------|--------|------------------------------------------------------------------------------------------------------------------------|

# STAGING- 5 STAGES OF CHRONIC KIDNEY DISEASE

|                                   | Stage 1                                                                                                      | Stage 2                                                                                    | Stage 3a                                                                                  | Stage 3b | Stage 4                                                                                           | Stage 5                                                                                                           |
|-----------------------------------|--------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|----------|---------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| Kidney function                   | 90%                                                                                                          | 60 - 89%                                                                                   | 45-59%                                                                                    | 30-44%   | 15 - 29%                                                                                          | <15%                                                                                                              |
| Symptoms                          | Normal kidney function but urine findings, structural abnormalities or genetic trait point to kidney disease | Mildly reduced kidney function and other findings (as for stage 1) point to kidney disease | Moderately reduced kidney function. Symptoms may include fatigue, poor appetite & itching |          | Severely reduced kidney function. Stage 3 symptoms get worse. Signs of edema in lower extremities | Very severe (end stage) kidney failure. Symptoms include poor sleep, itchiness, difficulty breathing and vomiting |
| eGFR<br>ml/min/1.73m <sup>2</sup> | ≥90                                                                                                          | 60 – 89                                                                                    | 45-59                                                                                     | 30-44    | 15 – 29                                                                                           | <15                                                                                                               |

Adapted from: KDOQI Clinical Practice Guidelines for Chronic Kidney Disease: Evaluation, Classification, and Stratification. National Kidney Foundation



# CURRENT KDIGO CLASSIFICATION OF CKD

## KDIGO

Prognosis of CKD by GFR and albuminuria categories

Low risk

Moderately increased risk

High risk

Very high risk

| Persistent albuminuria categories |                             |                          |
|-----------------------------------|-----------------------------|--------------------------|
| Description and range             |                             |                          |
| A1                                | A2                          | A3                       |
| Normal to mildly increased        | Moderately increased        | Severely increased       |
| <30 mg/g<br><3 mg/mmol            | 30-300 mg/g<br>3-30 mg/mmol | >300 mg/g<br>>30 mg/mmol |
|                                   |                             |                          |
|                                   |                             |                          |
|                                   |                             |                          |
|                                   |                             |                          |
|                                   |                             |                          |
|                                   |                             |                          |

|                                                             |     |                                  |       |
|-------------------------------------------------------------|-----|----------------------------------|-------|
| GFR categories<br>(ml/min/1.73 m²)<br>Description and range | G1  | Normal or high                   | ≥ 90  |
|                                                             | G2  | Mildly decreased                 | 60-89 |
|                                                             | G3a | Mildly to moderately decreased   | 45-59 |
|                                                             | G3b | Moderately to severely decreased | 30-44 |
|                                                             | G4  | Severely decreased               | 15-29 |
|                                                             | G5  | Kidney failure                   | <15   |



# PATHOPHYSIOLOGY

FYI

↓ Na reabsorption,  
↑ K, NH<sub>4</sub><sup>+</sup>, Ca/Phos  
metabolism altered

Loss of nephron mass

Remaining nephrons hypertrophy to compensate

Glomerular capillary hypertension

Systemic hypertension (mediated by ATII)

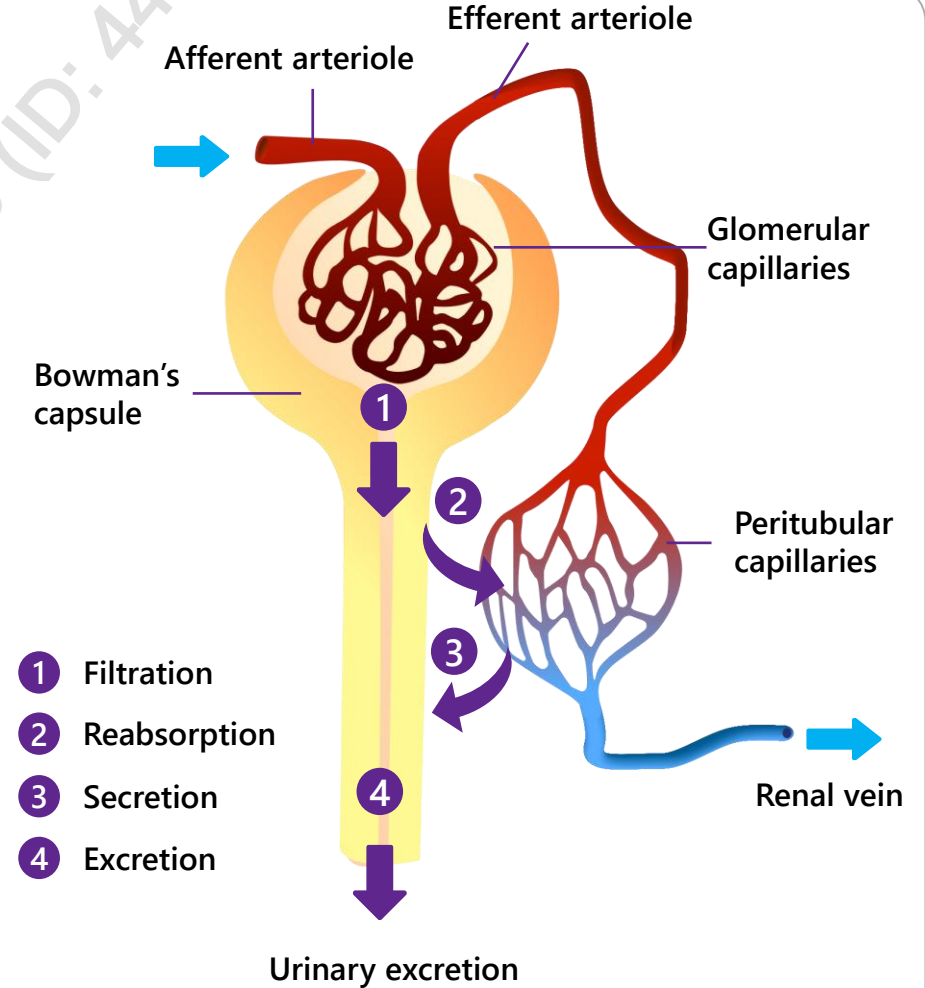
Impaired glomerular permeability

Albuminuria & proteinuria

Inflammation & cell damage

Progressive loss of nephron

Reduced glomerular filtration rate (GFR)



Excretion = Filtration - Reabsorption + Secretion

# CLINICAL PRESENTATION

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- Poor sleep/Fatigue/Drowsiness
- Shortness of breath
- Decreased appetite
- Nausea/Vomiting
- Constipation
- Leg swelling
- Muscle cramps
- Heartburn/Peptic ulceration
- Pruritis
- Sexual dysfunction
- Pain/Bone/joint pain
- Weakness
- Poor mobility
- Abnormal homeostasis
- Cold intolerance
- Malaise
- Anemia
- Neuropathy
- CHF
- Jaundice
- Seizures
- Decreased consciousness
- Renal osteodystrophy

# RISK FACTORS

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## Progressive Risk Factors

- **Uncontrolled diabetes**
  - Up to 1/2 of type II DM are at risk for Diabetic Nephropathy
- **Uncontrolled hypertension**
- **Autoimmune disorder**
  - Glomerulonephritis
  - Lupus nephritis
  - IgA nephropathy
- **CVD (incl heart failure)**
- **Smoking**
- **Obesity**

## Susceptibility Factors

- Advanced age
- Low socioeconomic status
- Low birth weight
- Family history of CKD
- Certain ethnic groups (likely due to access disparities and SDOH)
- Exposure to nephrotoxins.
- Drug-induced nephrotoxicity
- Prior AKI/AKD
- Kidney failure
- History of kidney infections & obstructions (kidney stones)
- Reduced kidney mass
- Polycystic kidney disease
- Systemic inflammation (e.g., HIV nephropathy)
- Dyslipidemia

# MEDICAL/DRUG RELATED CAUSES

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## Nephrotoxins (not an exhaustive list)

- IV contrast media agents used for angiograms
- Aminoglycosides, vancomycin
- Cisplatin
- Non-steroidal anti-inflammatory agents (NSAIDs)
- Diuretics
- ACE inhibitors/angiotensin receptor blockers (ARB) and direct renin inhibitors

## Phosphate containing products

- Can worsen the phosphate/calcium imbalance (e.g. fleet enema or other laxatives)

# DIAGNOSIS

Estimate GFR for staging of CKD

Investigate for renal structural abnormalities

- Biopsy, renal angiogram, ultrasound

Quantification of protein or microalbumin

- Urinalysis
- 24-hr urine collection
- Urine albumin: creatinine ratio [ACR] OR protein: creatinine ratios [PCR]

Blood chemistry for electrolyte and metabolic abnormalities

- Cr, Urea (BUN), Na, K, Phosphate, Ca (ionized), pH, Vitamin D, PTH, Hb, Platelet, Iron status (% sat, ferritin, serum iron)

Guidelines recommend testing people at risk for/with CKD using both urine albumin measurement and assessment of GFR

# PROTEINURIA

Proteinuria (aka albuminuria): elevated protein in the urine

- No proteinuria → microalbuminuria → macroalbuminuria → declining kidney function

| Stage                     | Evaluation and monitoring                                                                                                                                                                  | Treatment                                                                                                                                                                                                                                                          |
|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| No proteinuria            | <ul style="list-style-type: none"><li>• Monitor BP, BG and lipids</li><li>• Screen for microalbuminuria</li></ul>                                                                          | <ul style="list-style-type: none"><li>• BP medications to achieve BP targets, (Use ACEI or ARB preferentially)</li><li>• Dietary advice for sugar and fat</li><li>• Smoking cessation, Diabetes control</li><li>• CVD risk assessment for statin therapy</li></ul> |
| Microalbuminuria          | <ul style="list-style-type: none"><li>• Close monitoring of BP, BG and lipids</li><li>• Monitor urinary proteins &amp; SCr, CrCl</li></ul>                                                 | <ul style="list-style-type: none"><li>• All of the above</li><li>• Titrate BP &amp; BG medications to achieve targets</li><li>• Assess need for statin therapy</li></ul>                                                                                           |
| Macroalbuminuria          | <ul style="list-style-type: none"><li>• Close monitoring of BP, BG and lipids</li><li>• and complications (P, K, Ca, PTH, Hb)</li><li>• Monitor urinary proteins &amp; SCr, CrCl</li></ul> | <ul style="list-style-type: none"><li>• All of the above</li><li>• Aggressive dietary restriction (P, K, protein)</li><li>• Treat complications (anemia, hyperphosphatemia, hypocalcemia, hyperparathyroidism, gout, hyperkalemia)</li></ul>                       |
| Declining kidney function | <ul style="list-style-type: none"><li>• Close monitoring of BP, BG and lipids</li><li>• and complications (P, K, Ca, PTH, Hb)</li></ul>                                                    | <ul style="list-style-type: none"><li>• Same as above</li><li>• Prepare for dialysis and/or transplant</li></ul>                                                                                                                                                   |

# GOALS OF THERAPY

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- ✓ Treat reversible causes of renal dysfunction - hypovolemia, medications, obstruction
- ✓ Prevent or delay progression of CKD - BP, lipids and diabetes control
- ✓ Treat complications of CKD - anemia, hyperkalemia, hyperphosphatemia, hypocalcemia, hyperuricemia
- ✓ Prevent mortality related to cardiovascular complications in CKD
  - e.g. left ventricular hypertrophy, CHF, MI, sudden cardiac death
- ✓ Minimize side effects related to therapy by dose adjusting medications
- ✓ Avoid use of nephrotoxic agents



# NON-PHARMACOLOGICAL MEASURES

---

Smoking cessation

Healthy diet, physical activity, weight management to achieve optimal BMI (18.5-24.9)

Provide support to prevent depression

Parathyroidectomy → Last option for hyperparathyroidism

Assess for kidney & pancreas transplant (typically Stage 4 + Type 1 Diabetes)

# NON-PHARMACOLOGICAL MEASURES: DIET AND NUTRITION

---

## Sugar, carbohydrates

- Achieve A1C target (<7% for most patients, range 6.5-8%)

## Sodium intake

- Achieve desired SBP target <120mm Hg (<2g Sodium/day)

## Dietary potassium

- Avoid and minimize hyperkalemia

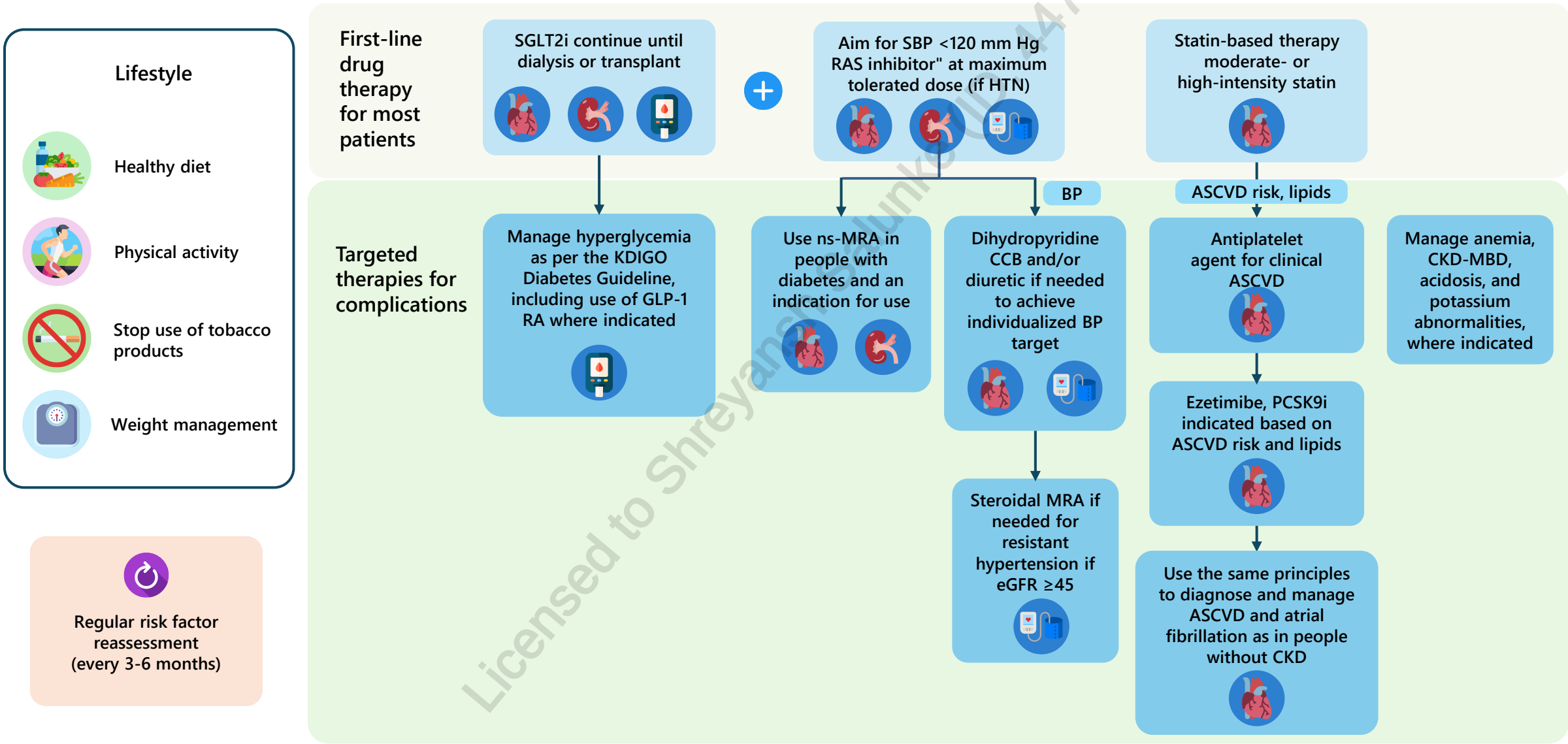
## Dietary phosphate (for >stage 3)

- Meats, dairy product, nuts, peanut butter, beer, whole grain, vegetables (broccoli, mushrooms, corn, peas)

## Dietary protein

- Aim for 0.8 g/kg/day in adults with CKD G3-G5

# COMPREHENSIVE TREATMENT



# PHARMACOLOGICAL MEASURES TO SLOW THE PROGRESSION OF CKD

## Blood pressure control

- Treat to target SBP <120 mmHg when tolerated
- **First-line: ACE/ARB when albuminuria present**; 2<sup>nd</sup> line: DHP CCB (if diabetic + renal disease/microalbuminuria) or diuretic. T2DM? Think Finerenone!
- Monitor K, SCr, GFR within 2 weeks of initiation or dose increase of ACEI or ARB (earlier in high-risk group)

## Target urine ACR < 3 mg/mmol or PCR < 50 mg/mmol

- Use **ACEI or ARB as 1<sup>st</sup> line for albuminuria or proteinuria**
- ACEI/ARB → most effective at preserving renal function by preferential dilation of efferent arterioles, thereby decreasing intraglomerular capillary pressure
- ACE/ARB should be held in cases of severe dehydrations (Sick Day Management)
- ACE/ARB: monitor for increase in K<sup>+</sup>, angioedema, dry cough (ACEI only)

## Blood glucose control (T2DM)

- **A1C target 6.5-8% (<7% for most patients)**
- **First-line: metformin, SGLT2i (eGFR ≥ 20 mL/min/1.73 m<sup>2</sup> or heart failure), 2<sup>nd</sup> Line: GLP-1RA**
- **Avoid metformin if eGFR < 15-30 mL/min/1.73 m<sup>2</sup>\***
- Monitor for hypoglycemia
- If blood sugars not met, add GLP-1 RA

\* Metformin is contraindicated if eGFR is <15 mL/min/1.73 m<sup>2</sup>, but this may vary based on condition e.g. in chronic kidney disease (CKD), if eGFR is <30 mL/min/1.73 m<sup>2</sup>, metformin must be stopped at the time of or before the administration of iodinated contrast media due to the risk of lactic acidosis

# PHARMACOLOGICAL MEASURES TO SLOW THE PROGRESSION OF CKD

## Implement measures to modify CVD risk factors

- Screen and calculate CVD risk (high risk population as per dyslipidemia guidelines)
- Oral low-dose aspirin to prevent recurrent ischemic CVD events if established ischemic CVD
- Treat hyperlipidemia, target LDL < 2.0 mmol/L: use statin and/or ezetimibe therapy
  - Exception: do not initiate in dialysis patients

## Avoid nephrotoxins

- Aminoglycosides, NSAIDs, COX-2 inhibitors, contrast

## Vaccination

- Influenza annually; pneumococcal vaccine (may need booster in some populations)
- Patients with CKD have decreased immunity

## Treat complications

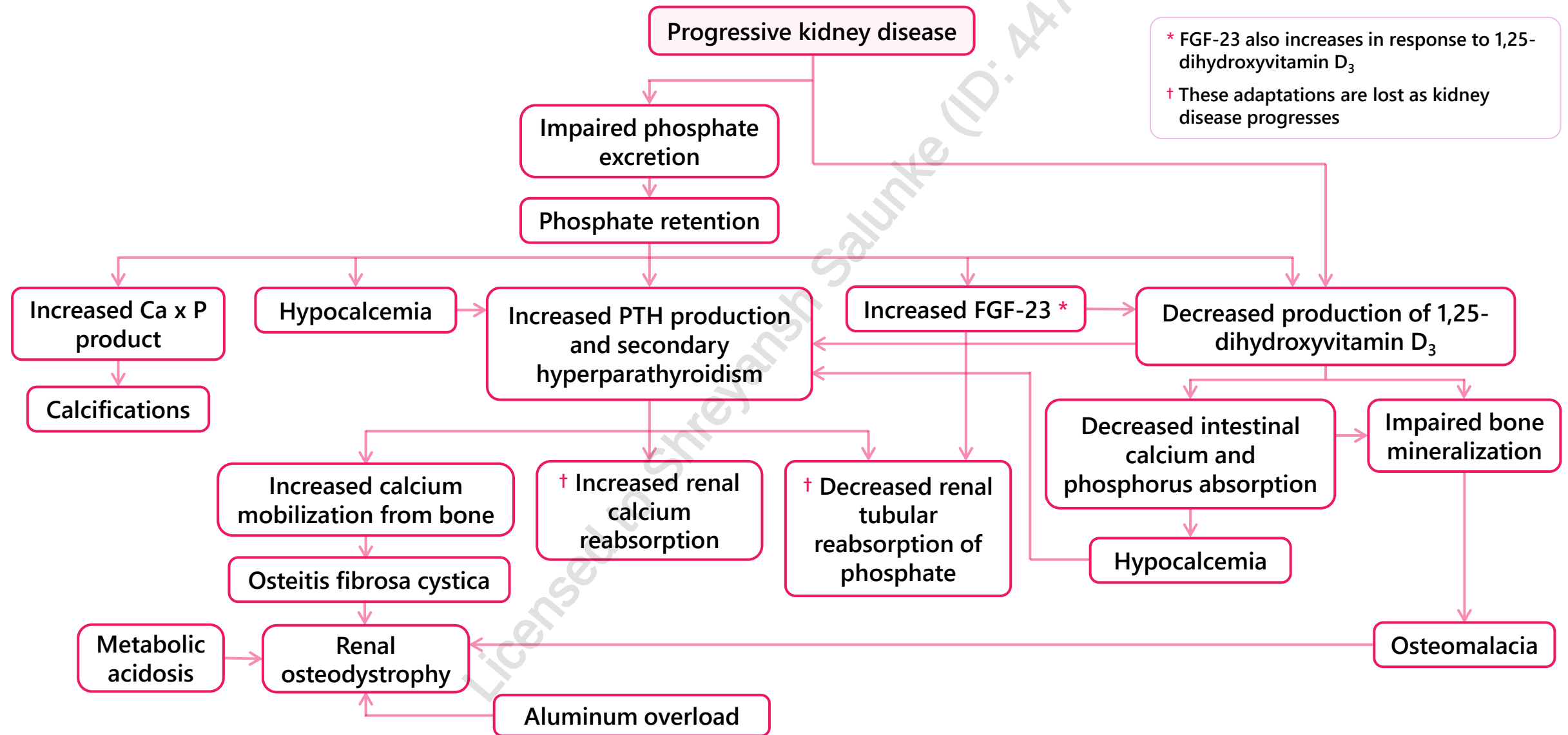
- Anemia: ↓ erythropoietin production
- Hyperphosphatemia: ↓ excretion of phosphate by kidney
- Secondary Hyperparathyroidism: ↑ phosphate, ↓ calcium and ↓ active Vitamin D

# COMPLICATIONS OF CKD: OVERVIEW

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- Hyperphosphatemia
- Secondary Hyperparathyroidism
- Anemia and Iron Deficiency

# COMPLICATIONS OF CKD







# HYPERPHOSPHATEMIA

## Normal Kidney Function

- Eliminates P and reabsorbs Ca (impaired kidney function causes  $\uparrow$  P and  $\downarrow$  Ca)

## Consequences of Hyperphosphatemia

- Pruritus, extra-skeletal calcification (Ca and P precipitates into blood vessels, heart, soft tissue, other organs), cardiac complications, bone disease (indirectly by  $\uparrow$  PTH),  $\uparrow$  mortality and morbidity

## Treatment

- Diet: restrict to 800-1200 mg/day, Reduce phosphate-processed foods & not healthy/nutritious value foods. Refer patient to dietician.
- Dialysis: HD removes 900 mg phosphorus/treatment whereas PD removes 300 mg each day
- Phosphate binders (bind to dietary P in the intestine to form an insoluble complex that cannot be absorbed in the bloodstream resulting in P removal in the stool):
  - Take with meals to prevent P absorption
  - Interacts with Ca, iron, levothyroxine, quinolones, tetracyclines, alendronate, antacids
  - Side effects: GI intolerance; aluminum-based binders can cause dementia, worsening anemia, improper bone mineralisation

# PHOSPHATE (P) BINDERS

|                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Calcium based phosphate binders          | <p>Calcium carbonate (Tums®) (40% elemental Ca<sup>2+</sup>) , Calcium citrate (20% elemental Ca<sup>2+</sup>)</p> <ul style="list-style-type: none"><li>• <b>Side effects:</b> Constipation, Nausea, Hypercalcemia - especially with vit D OR with decreased bone turnover (osteomalacia)</li><li>• <b>Interactions:</b> oral iron salts, fluoroquinolones, tetracyclines, levothyroxine, H2 blockers and PPI increase gastric pH and reduce dissolution of Ca based Phosphate binders (except Ca citrate which is less dependent on acidic gastric pH for dissolution)</li></ul>                                                                                 |
| Aluminum-based phosphate binders (AVOID) | <ul style="list-style-type: none"><li>• Aluminum salts</li><li>• Potent but high doses required which can cause aluminum toxicity</li><li>• <b>Not recommended by US and Global guidelines</b></li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Non-calcium containing binders           | <p>First-line agent in those with hypercalcemia; reduces vascular calcification</p> <ul style="list-style-type: none"><li>• <b>Sevelamer carbonate (Renvela®) and hydrochloride (Renagel®):</b> Taken with meals, can worsen metabolic acidosis (with hydrochloride formulation), has minimal GI absorption</li><li>• <b>Lanthanum:</b> large chewable pills (need to chew completely), taken with meals; separate other drug dosing by 2 hours to minimize interactions</li><li>• <b>Sucroferric oxyhydroxide (Velporo®):</b> new agent containing 500 mg of iron/tablet (insignificant absorption); taken with meals, not recommended in iron overload</li></ul> |

# SUMMARY OF PHOSPHATE BINDERS

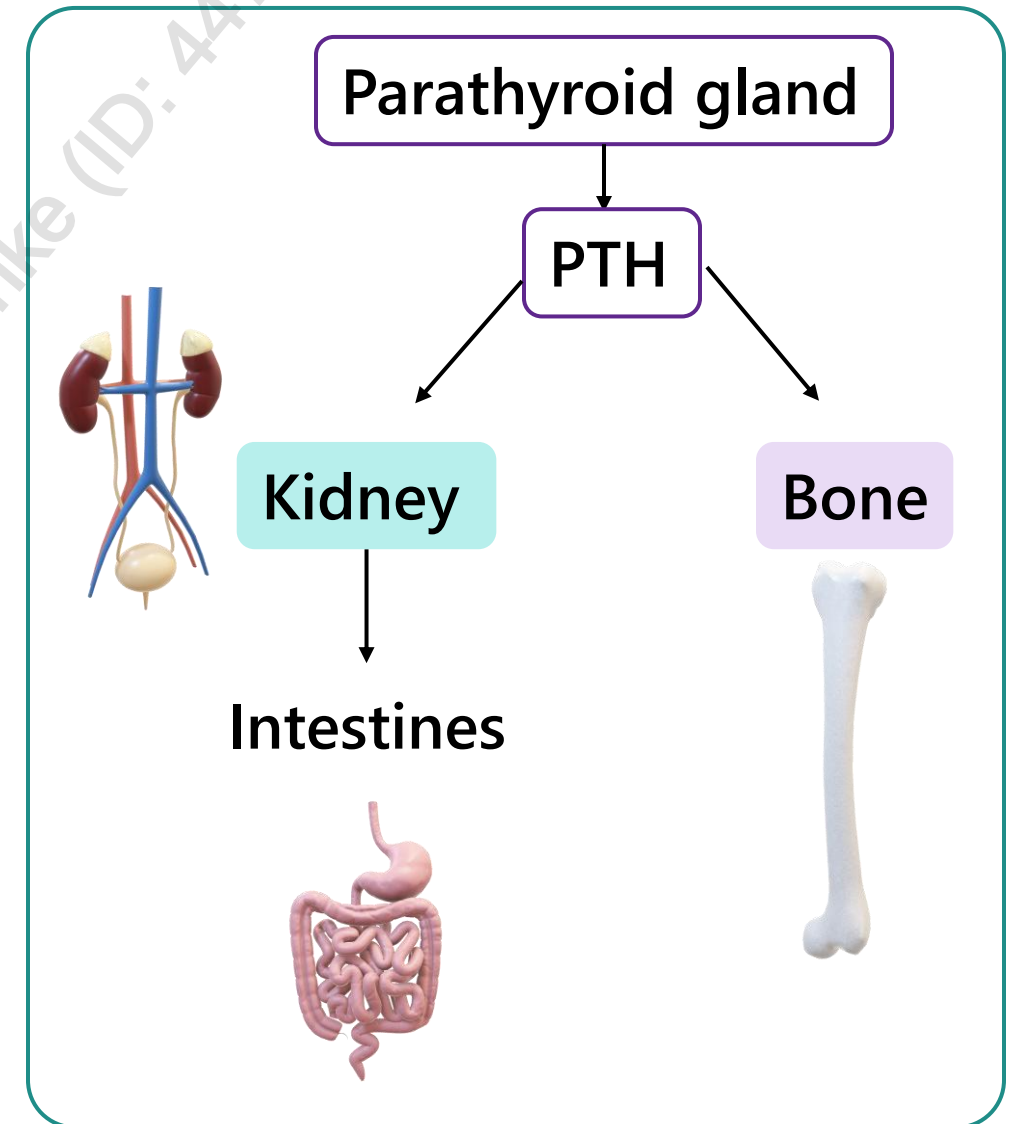
| Phosphate binder                     | Advantages                                                | Disadvantages                                                   |
|--------------------------------------|-----------------------------------------------------------|-----------------------------------------------------------------|
| Calcium carbonate or calcium acetate | Effective                                                 | Pill burden, GI side effects, hypercalcemia (less with acetate) |
| Aluminum salts                       | High efficacy                                             | Aluminum toxicity, no safe dose                                 |
| Sevelamer                            | Calcium free, effective, ↓total and LDL cholesterol       | GI side effects, pill burden                                    |
| Lanthanum                            | Calcium free, ↓pill burden, unaffected by gastric acidity | GI side effects                                                 |
| Sucroferric Oxyhydroxide             | Calcium free<br>Unaffected by gastric acidity             | Discoloration of stool<br>Diarrhea with initial treatment       |

- Calcium-based phosphate binders are first choice because of convenience and cost
- Non calcium containing binders (mostly sevelamer or lanthanum) used in patients with hypercalcemia or as an adjunct
- When choosing a phosphate binder, consider compliance, cost, side effects

# SECONDARY HYPERPARATHYROIDISM

## Normal Function of PTH:

- Parathyroid gland secretes parathyroid hormone (PTH)
- PTH acts on the kidney & bone to regulate Ca and  $\text{PO}_4$  levels
- PTH increases Ca by:
  - Stimulating resorption of Ca from bone
  - Increased Ca reabsorption by kidneys
  - Stimulating kidney to metabolize vitamin D precursors to active vitamin D (calcitriol) → calcitriol ↑ intestinal absorption of Ca and  $\text{PO}_4$
- PTH regulates P by:
  - Stimulating resorption of P from bone
  - Decreased reabsorption of P from kidneys



# SECONDARY HYPERPARATHYROIDISM

## ↓Ca

- Insufficient Ca to bind to Ca sensing receptors expressed in parathyroid glands

## ↓Ca-sensing Receptors

- - ↓ sensitivity to calcium levels
- - Higher serum calcium to suppress secretion of PTH

## ↑PO<sub>4</sub>

- Exact mechanism unknown
- Interfere with level of ionized Ca

## ↓ active Vit. D

- ↓intestinal Ca absorption
- Acts on vitamin D receptors in parathyroid gland (low calcitriol)

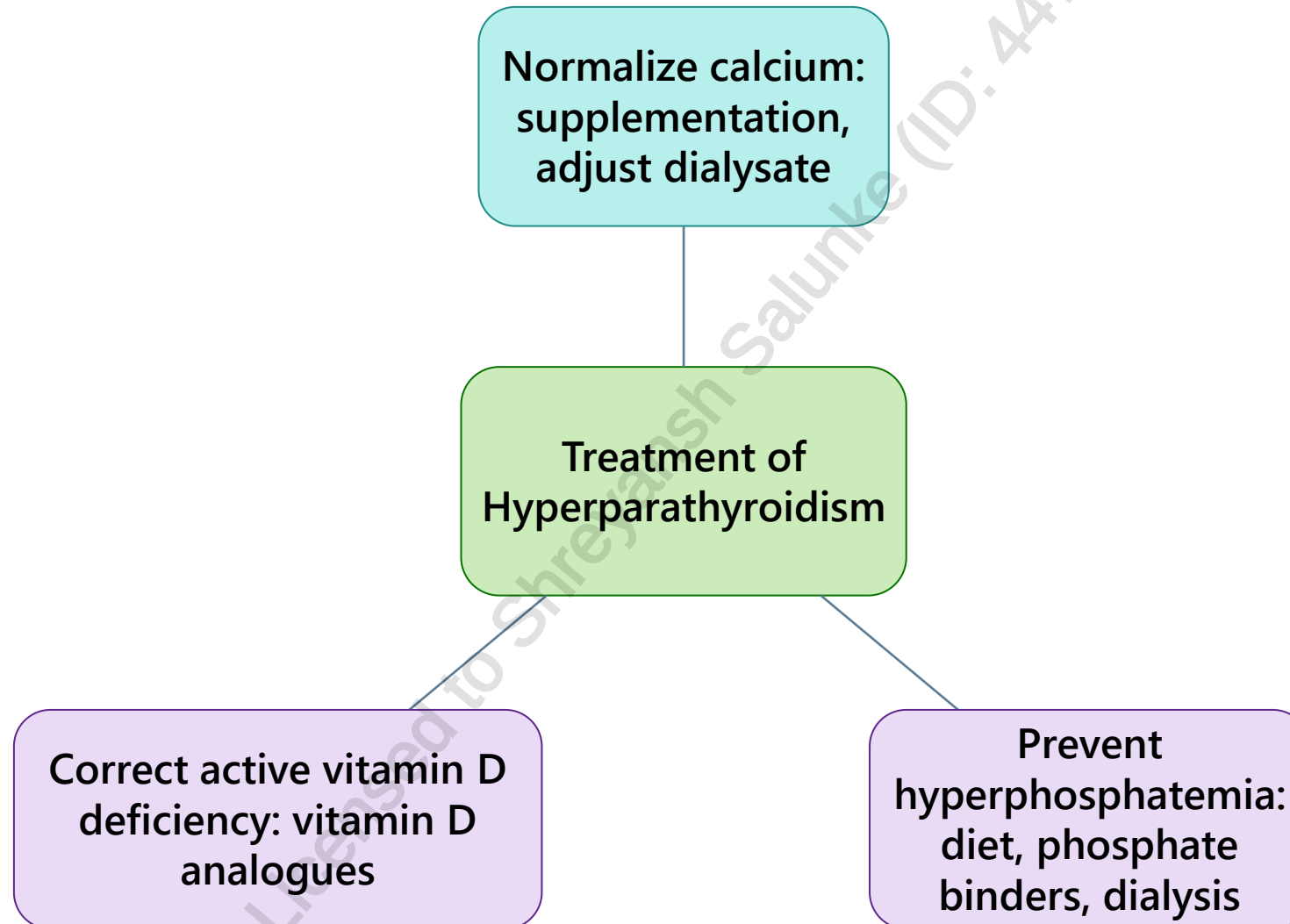
## Consequences

- High turnover bone disease
- Osteitis fibrosa cystica: ↑ fracture risk
  - ↑ PTH stimulate osteoclasts & osteoblasts
  - ↑ resorption and bone formation
  - Results in abnormal unmineralized osteoid
  - Bone is brittle and ↓ strength

Constant stimulation by ↑P and ↓Ca causes hyperplasia of the parathyroid gland ↑PTH production:  
→ PTH secretion correlated to size of gland

# TREATMENT OF HYPERPARATHYROIDISM

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# VITAMIN D ANALOGUES

|                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| MOA                          | <ul style="list-style-type: none"><li>• Stimulates calcium absorption in intestines</li><li>• Stimulates renal tubular calcium reabsorption</li><li>• Suppresses PTH production</li><li>• Causes bone remineralization</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Analogues types and comments | <ul style="list-style-type: none"><li>• Alfacalcidol PO (One-Alpha ®), calcitriol PO/IV (Rocaltrol ®)</li><li>• <b>No conclusive studies to support use of a specific derivative over another</b></li><li>• Indicated for treatment of secondary HPT &amp; hypocalcaemia</li><li>• IV more practical in hemodialysis patients but is more expensive than PO</li><li>• <b>There is no difference in efficacy between IV and PO route</b></li><li>• According to the guidelines: calcitriol &amp; vitamin D analogs not to be commonly used. Retained for patients with CKD G4-G5 [eGFR &lt;30 min/1.72 m<sup>2</sup>] with hyperparathyroidism that is progressive and critical in nature .</li></ul> |
| Precautions                  | <ul style="list-style-type: none"><li>• Over- suppression of PTH: adynamic bone disease</li><li>• Decreased osteoblast and osteoclast activity; calcium uptake by bone is reduced; increased risk of fracture and calcification</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Side effects                 | <ul style="list-style-type: none"><li>• <b>Hypercalcemia, hyperphosphatemia</b></li><li>• <b>↑ Ca × P product → leading to soft tissue and vascular calcification</b></li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

# CALCIUM MIMETICS FOR HYPERPARATHYROIDISM (HPT)- CINACALCET

|              |                                                                                                                                                                                                                                                      |
|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| MOA          | <ul style="list-style-type: none"><li>• Binds allosterically to Ca receptors to increase sensitivity to Ca and decrease PTH release</li><li>• No increase in calcium or phosphate</li><li>• Significantly decreases PTH and Ca x P product</li></ul> |
| Analogues    | <ul style="list-style-type: none"><li>• Take with meals to increase bioavailability</li><li>• Titrate every 2-4 weeks (severe hypocalcemia can occur if not slowly titrated)</li><li>• Metabolized by CYP3A4, 2D6, 1A2 and inhibits CYP2D6</li></ul> |
| Comments     | <ul style="list-style-type: none"><li>• Over- suppression of PTH: adynamic bone disease</li><li>• Decreased osteoblast and osteoclast activity; calcium uptake by bone is reduced; increased risk of fracture and calcification</li></ul>            |
| Side effects | <ul style="list-style-type: none"><li>• Hypocalcemia: measure calcium after 1 week then every 3 weeks</li><li>• Nausea, vomiting (transient and mild)</li></ul>                                                                                      |



# ANEMIA AND IRON DEFICIENCY

Normal Kidney Function: Produces erythropoietin

## Impaired kidney Function

- ↓ epoetin production leading to ↓ RBC
- Decreased RBC life span due to uremia (120 → 60 days in ESRD)
- Iron deficiency, blood loss due to blood tests & hemodialysis

## Consequences

- Anemia
- Increased cardiac output & left ventricular hypertrophy
- **Reduced oxygen utilization**  
→ Fatigue, SOB
- **↑ mortality**
- **↑ hospitalization**
- **↑ progression to ESRD**
- **↑ depression**
- **↑ morbidity**
- **↓ QOL**

## Diagnosis

- Rule out other causes of anemia (ex. GI bleed, Vitamin B12 or folate deficiency)

### Anemia:

- Males: Hb <130g/L
- Non-pregnant females: Hb <120g/L
- Iron studies: CBC, ferritin, TSAT, TIBC, serum iron

Treatment options: **Iron replacement (usually trialled 1<sup>st</sup>) and/or Epoetin stimulating agents (ESA)**

# IRON DEFICIENCY IN CKD

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Caused by ESA, GI bleeding, iron deficiency, CKD

- Treatment initiated when ferritin  $\leq 500$  ng/mL AND TSAT  $\leq 30\%$
- **Treatment: iron supplementation**
  - Alternative to ESA to produce modest  $\uparrow$  in Hb
  - Boost iron stores prior to initiating ESA therapy
  - Adjuvant to ESA to increase ESA response (up to 30-75% decrease in ESA requirement)
  - Treatment of iron deficiency in patients on ESA therapy
- **Administration route: PO, IV, IM**
  - Oral  $\rightarrow$  GI intolerance (dyspepsia, constipation), poor compliance
  - IV  $\rightarrow$  more effective than other routes but requires monitoring to avoid hypersensitivity reactions
  - IM  $\rightarrow$  painful and causes skin discolouration, least common route

# ORAL IRON SUPPLEMENTATION

| Iron                | Dose   | Elemental iron |
|---------------------|--------|----------------|
| Iron polysaccharide | 150 mg | 150 mg         |
| Ferrous fumarate    | 300 mg | 100 mg         |
| Ferrous sulfate     | 300 mg | 60 mg          |
| Ferrous gluconate   | 300mg  | 35 mg          |

- Advise patients to take **iron on empty stomach** to increase absorption
- **Vitamin C** can enhance absorption of iron salts
- Take iron **1 hour before or 2-3 hours after dairy products/Ca supplements**
- Warn patients about **dark stools, constipation, abdominal pain (with higher elemental iron intake)**

# IV IRON SUPPLEMENTATION

Use IV iron if: patient is unable to tolerate iron, unable to reach target while on PO iron; wait at least 2 weeks after loading dose to reassess iron indices

With all formulations, monitor for: **anaphylactoid reactions, hypotension** (avoid rapid IV infusion)  
Other side effects: **fever, chills, headache, myalgia, arthralgia, urticaria, dizziness, anaphylaxis** (rare)

| Formulation                                     | Comments                                                                                                                                                                                                                                                     |
|-------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Iron sucrose<br>(Venofer®)                      | <ul style="list-style-type: none"><li>• Most data in patients on hemodialysis</li><li>• 100 mg 1-3 times weekly IV infusion, bolus can be given in 2 minutes or as short infusions (2 hours)</li><li>• Minimal side effects if used at lower doses</li></ul> |
| Iron isomaltoside<br>(Monoferic®)               | <ul style="list-style-type: none"><li>• 100 mg 1-3 times weekly IV infusion or Bolus</li></ul>                                                                                                                                                               |
| Sodium ferric gluconate complex<br>(Ferrlecit®) | <ul style="list-style-type: none"><li>• 125 mg administered IV infusion 1-3 times weekly</li><li>• Higher doses (above 125mg) associated with hypotension, acute onset diarrhea</li></ul>                                                                    |

# ERYTHROPOIESIS-STIMULATING AGENTS (ESAS)

- Use ESA to keep **target Hb <115g/L**
- Generally, initiate when Hb 90 - 100g/L
- Available as IV and SC
- **Epoetin alpha** (Eprex®)
  - 1<sup>st</sup> recombinant human erythropoietin launched in the market
  - Shorter half-life
  - Administer 1-3 times/week to monthly
- **Darbepoetin alpha** (Aranesp®)
  - Synthetic erythropoietin analogue
  - Longer half-life & higher potency compared to epoetin alpha
  - Administer every 1-2 weeks to monthly
  - Less immunogenic than epoetin alpha

## Issues with ESA:

- **Thromboembolism** (with higher Hb targets)
- **Hypertension** (due to increase vascular resistance)
- Allergic reactions (epoetin alpha)
- Myalgia, flulike symptoms
- Injection site pain
- Nausea, vomiting
- **ESA complications**
  - Antibody production
  - Aluminum toxicity
  - Pure red cell aplasia

# MONITORING PARAMETERS- THERAPEUTIC ENDPOINTS

| Parameter                | Degree of Change                           | Timeframe |
|--------------------------|--------------------------------------------|-----------|
| PO <sub>4</sub>          | Within normal limits                       | 3 months  |
| Ca x P Product           | Prevent                                    | 3 months  |
| PTH                      | Avoid progressive rising                   | Ongoing   |
| HgB                      | Attain target Hb (but don't exceed 115g/L) | 3 months  |
| Ferritin                 | Target > 500 ng/mL                         | 3 months  |
| BP                       | Decrease SBP <120 mmHg                     | 3 months  |
| Albuminemia, Proteinuria | ACR <3 mg/mmol, PCR <50 mg/mmol            | 3 months  |
| HbA1c                    | Attain A1C target                          | 3 months  |
| Weight                   | Maintain target BMI                        | Ongoing   |
| Dyslipidemia (LDL)       | LDL < 2 mmol/L                             | 3 months  |

# MONITORING PARAMETERS- SAFETY ENDPOINTS

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| Parameter                | Degree of Change | Timeframe                              |
|--------------------------|------------------|----------------------------------------|
| Hypercalcemia            | Prevent          | While on Phosphate binder therapy      |
| Malnutrition             | Prevent          | While on dietary phosphate restriction |
| Cough, angioedema        | Prevent          | While on ACEI therapy                  |
| Hypoglycemia             | Prevent          | While on T2DM therapy                  |
| Myopathy                 | Prevent          | While on statin therapy                |
| GI side effects          | Prevent-minimal  | While on oral iron, phosphate binders  |
| Injection site reactions | Minimal          | While on ESA, IV iron therapy          |

# THERAPEUTIC TIPS

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## Sick day management:

Primary concern is **dehydration** [example: patient is vomiting and has diarrhea but is not consuming adequate fluids]

- **Step 1:** Encourage patient to consume fluids and electrolytes.
- **Step 2:** Advise patient to check sugars frequently and watch out for hypoglycemia & hyperglycemia/ signs of DKA
- **Step 3:** Keep basal insulin. Bolus insulin adjusted on the advice of a diabetes educator.
- **Step 4:** Hold certain medications until safe to resume:
  - **S:** Sulfonylureas
  - **A:** ACE inhibitors
  - **D:** Diuretics
  - **M:** Metformin
  - **A:** ARBs
  - **N:** Nonsteroidal anti-inflammatory drugs
  - **S:** SGLT2 inhibitors



# THERAPEUTIC TIPS

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- **Dose adjust common medications** in CKD (list not inclusive)
  - Penicillin's, quinolones, allopurinol, gabapentin, pregabalin, sulfonamides, colchicine, tramacet (acetaminophen/tramadol)
- **Avoid nephrotoxins** (list not inclusive)
  - NSAIDs (include COX2 inhibitors), vancomycin, aminoglycosides, ACEI, ARBs, diuretics
- **Avoid certain drugs due to lack of efficacy in CKD** (list not inclusive)
  - Thiazide diuretics & Nitrofurantoin
- **Avoid drugs due to side effects related to accumulation** (list not inclusive)
  - Glyburide (due to hypoglycemia – instead, use gliclazide)
  - Long-acting opioids (due to excessive drowsiness - use short-acting instead)
  - Meperidine (due to risk of seizures)
- Prevent & treat diabetic nephropathy

# TAKEAWAY

## Functions of the Kidney

### Excretory

- Elimination of waste products e.g. urea, Cr, uric acid
- Water
- Electrolytes e.g. Na, K, Cl, H

### Endocrine

- Produces 1, 25-dihydroxcholecalciferol
- Renin
- Erythropoietin
- Prostaglandins

### Regulatory and metabolic

- Fluid Balance
- BP
- Tubular Re-absorption e.g. glucose, amino acids, bicarbonate
- Bone density (metabolic)

What happens  
When it stops working?

## Clinical Consequences of Renal Failure

1. Urinary Symptoms (loss of ability to concentrate urine)
2. Uremia – anorexia, vomiting, constipation, foul taste, skin discolour, itchiness
3. Fluid retention – as GFR ↓ body is unable to excrete salt and water adequately.
  - Edema (periphery, pulmonary, ascites, sacral, eyes on waking)
4. Proteinuria – due to leaky glomerulus
5. Electrolyte Disturbances (hyperphosphatemia, hyperparathyroidism)
  - Na (Hypo/hyper)
  - K (High)
  - H ions (accumulation --> acidosis)
  - PO4 (high)
  - Ca (high/low)
6. Anemia – due to inadequate secretion of erythropoietin. Fatigue, breathlessness, increased in cardiac output --> tachycardia, palpitations, heart failure.
7. Hypertension
8. Renal Osteodystrophy
  - secondary hyperparathyroidism
  - vitamin D Deficiency
9. Gastrointestinal – constipation, nausea

What drugs are used to  
Control these?

## Commonly used drugs

1. Antihypertensives. E.g. ACEI, angiotensin 2 receptor antagonists, beta blockers.
2. Diuretics – manage fluid retention with loop diuretics (high dose) and fluid restriction
3. Phosphate binders – to lower phosphate e.g. sevelamer
4. Vitamin supplementation as water soluble vitamins may become lost during dialysis.
5. Laxatives – to treat severe constipation caused by fluid restriction, phosphate binders.
6. Synthetic erythropoietin agents, to stimulate red blood cell production, often with iron supplementations. Folate may need to be corrected.
7. Sodium bicarbonate – to correct acidosis.
8. Vitamin D analogues– to correct calcium level and manage hyperparathyroidism.
9. Cardiovascular risk reduction- e.g statins

# CASE 1(A)

KP is a 52-year-old female office manager. KP presents to your pharmacy with some questions. She has been experiencing fatigue, nausea and malaise for several months. It has been a couple of years since her last checkup with her family doctor, and at that time he said "her kidneys were slowing down". KP has a past medical history of hypertension, diabetes, chronic back pain, and obesity. She currently takes naproxen 500 mg po bid, hydromorphone cr 3 mg bid, hydromorphone 1-2 mg q4-6h prn, ramipril 5 mg once daily, hydrochlorothiazide 25 mg once daily, metformin 1000mg po bid, acetaminophen/methocarbamol 1-2 tablets once daily PRN, and ASA 81 mg once daily. She does not smoke or drink alcohol. She drinks 3-4 cans of caffeinated soda/day. She has an allergy to sulfamethoxazole/trimethoprim, resulting in swelling.

## Objective findings (3 years ago):

**Hb = 110**

(Normal: 115-155 g/L)

**WBC = 6.0**

(Normal:  $3.5-10.5 \times 10^9/L$ )

**Plt = 234**

(Normal:  $130-380 \times 10^9/L$ )

**Ca<sup>2+</sup> = 1.10**

(Normal: 1.14-1.28 mmol/L)

**ACR: 4.5**

(Normal:  $\leq 2.0$  g/mol Cr)

**BUN: 9**

(Normal: 2.1-8.0 mmol/L)

**SCr: 95**

(Normal: 22-75  $\mu\text{mol/L}$ )

**CrCl: 64**

**HbA1c: 8.1%**

(Normal: 4.8%-6.0%)

**Na = 139**

(Normal: 136-146 mmol/L)

**K<sup>+</sup>: 5.5**

(Normal: 3.5-5.1 mmol/L)

**PO<sub>4</sub>: 2.3**

(Normal: 0.81-1.58 mmol/L)

**BP: 175/72 mmHg**



# CASE 1(A) - QUESTIONS

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1. What is the BEST course of action?
  - a) Refer to ER immediately as she may have an AKI
  - b) Advise her to go home and rest as she may have the flu
  - c) Advise her to make an appointment with her family doctor as her CKD may be progressing
  - d) Recommend dimenhydrinate to help with the fatigue and nausea, follow up in 3 days
  
2. Which of the following of her medications are NOT considered nephrotoxic?
  - a) Naproxen
  - b) Ramipril
  - c) Hydromorphone CR
  - d) Hydrochlorothiazide

# CASE 1(B)

Two weeks later, KP has returned after seeing her doctor. He obtained new bloodwork and told her that “her kidneys have gotten worse”. He is planning to start her on a new medication to help “balance out her bloodwork” and has some questions for you. Her medications remain the same and you learn she has no private drug coverage.

## Objective findings:

BP = 143/87 mmHg

Hb = **103**

(Normal: 115-155 g/L)

Ferritin = **8**

(Normal: 11-307 µg/L)

tSat = **0.08%**

Normal: 0.20%-0.50%)

Corrected  $\text{Ca}^{2+}$  = 2.23

$\text{PO}_4^{2-}$  = **3.32**

(Normal: 0.81-1.58 mmol/L)

PTH = **17**

(Normal: 1.6-9.3 pmol/L)

HbA1c = **7.5%**

(Normal: 4.8%-6.0%)

SCr = **133**

(Normal: 22-75 µmol/L)

CrCl = **34ml/min**



# CASE 1(B) - QUESTIONS

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1. Which of the follow medications is the MOST appropriate one that KP should start?
  - a) Calcium carbonate
  - b) Lanthanum carbonate
  - c) Calcitriol
  - d) Cinacalcet
  
2. What is the BEST course of action for KP's anemia?
  - a) Start IV iron sucrose
  - b) Start oral iron supplementation
  - c) Start SC darbepoetin
  - d) Initiate a one time blood transfusion

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# CHANGE LOG

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## March 2025

- Slide 18: Added finerenone

## May 2025

- Slide 18: Added 'if diabetic + renal disease/microalbuminuria' next to CCB in accordance with HTN and KDIGO guidelines

## July 2025

- Content reviewed, minor formatting changes made

## November 2025

- Formatting, grammar changes
- Slide 3: Updated legend
- Slide 18: Updated information on metformin use in renal impairment and added footnote
- Slide 45: Updated references

## January 2026

- Minor formatting edits made throughout
- Slide 4: added mg/dL for Cockcroft-Gault
- Slide 5 – New
- Slide 11: removed race as risk factor
- Slide 19: added 2<sup>nd</sup> line for T2D and CKD (GLP-1RA)
- Slide 25: Removed Fosrenol® brand name as not available on the Canadian market anymore
- Slide 37: Edited ACR value to <3 mg/mmol